PHYSICS I / INSTRUCTOR ONLY

Diagnostic Key & Facilitation Guide

Keep outside student retrieval / professor-side only

Usage. Treat incorrect choices as potential evidence of misconceptions, not definitive labels. The agent may score multiple-choice items exactly, but short answers need criterion-based feedback and a human reviewer. No unvalidated percentage mastery claims.

Answer key

Q01 Answer: B

Zero net force means zero acceleration; the existing velocity remains constant.

Misconception hypothesis by distractor: **A:** MOTION_NEEDS_FORCE; **C:** FORCE_CAUSES_SPEED; **D:** FORCE_CAUSES_SPEED

Q02 Answer: C

Net force is 10 - 4 = 6 N right; a = 6/2 = 3.0 m/s² right.

Misconception hypothesis by distractor: **A:** NET_VS_APPLIED; **B:** NET_VS_APPLIED; **D:** NET_VS_APPLIED

Q03 Answer: B

For equal net forces, a = F_net/m; halving the mass doubles acceleration.

Misconception hypothesis by distractor: **A:** MASS_ACCEL; **C:** MASS_ACCEL; **D:** MASS_ACCEL

Q04 Answer: B

Earth acts on the book, and the book acts on Earth with an equal and opposite gravitational force. The forces act on different bodies.

Misconception hypothesis by distractor: **A:** FBD_PAIR; **C:** FBD_PAIR; **D:** FBD_PAIR

Q05 Answer: C

Constant velocity implies zero vertical acceleration, so vertical forces balance: N - mg = 0.

Misconception hypothesis by distractor: **A:** VELOCITY_FORCE; **B:** VELOCITY_FORCE; **D:** VELOCITY_FORCE

Q06 Answer: B

The box remains at rest, so net horizontal force is zero. Static friction adjusts to 12 N left as long as its maximum is not exceeded.

Misconception hypothesis by distractor: **A:** STATIC_FRICTION_ZERO; **C:** STATIC_FRICTION_MAX; **D:** FRICTION_WITH_MOTION

Q07 Answer: A

The interaction forces are a third-law pair: equal and opposite, acting on different bodies. Accelerations can differ due to mass.

Misconception hypothesis by distractor: **B:** THIRD_LAW_UNEQUAL; **C:** THIRD_LAW_UNEQUAL; **D:** FBD_PAIR

Q08 Answer: C

The component of gravity along the incline is mg sin 30°. Thus a = g sin 30° = 4.905 m/s² ≈ 4.91 m/s².

Misconception hypothesis by distractor: **A:** INCLINE_COMPONENT; **B:** INCLINE_COMPONENT; **D:** INCLINE_COMPONENT

Q09 Short-response criteria

A classmate says: "If a cart keeps moving, there must be a force pushing it forward." Explain whether this is correct for a cart moving at constant velocity on an ideal frictionless track. Then explain why everyday carts on rough floors often stop without being pushed.

- On a frictionless track, constant velocity requires zero net force, not a forward sustaining force.
- On a rough floor, friction is an opposing external force that produces deceleration.
- Distinguishes velocity from acceleration, and ideal model from everyday friction.

Review manually; ask a follow-up before assigning persistent learning labels.

Q10 Short-response criteria

Two blocks of masses 2.0 kg (A) and 3.0 kg (B) touch on a frictionless horizontal surface. A horizontal 10 N force pushes block A to the right. Find (i) the acceleration of the two-block system, (ii) the contact force of A on B, and (iii) the contact force of B on A. Name which two forces form a third-law pair.

- Total mass is 5.0 kg, so acceleration is $10/5 = 2.0\ m/s^2$ right.
- Block B has only the contact horizontal force: $F(A\text{ on }B) = 3.0 \times 2.0 = 6.0$ N right.
- $F(B\text{ on }A) = 6.0$ N left; this and $F(A\text{ on }B)$ are the third-law pair.
- Applied 10 N on A is not the reaction to $F(A\text{ on }B)$.

Review manually; ask a follow-up before assigning persistent learning labels.

Peer facilitation protocol

G01 - Motion without a push

Prompt: Discuss what would happen on an ideal frictionless surface and on a rough floor. Each student must use the phrase "net force" correctly.

Suggested AI move: Ask Student A to propose a test distinguishing constant velocity from acceleration. Invite Student B to challenge with a rough-floor counterexample. Avoid stating the complete answer first.

Possible trigger: MOTION_NEEDS_FORCE

G02 - Truck and car disagreement

Prompt: Label the two interacting bodies and identify the forces for each. Explain which mass matters for acceleration.

Suggested AI move: Ask each student to name the source and recipient of one force before accepting a conclusion.

Possible trigger: THIRD_LAW_UNEQUAL, FBD_PAIR

G03 - The static-friction trap

Prompt: A 12 N horizontal push does not move the box. Use Newton's second law to decide the actual friction force and explain why both statements can be wrong.

Suggested AI move: Request a horizontal free-body diagram and ask for net force before asking about maximum static friction.

Possible trigger: STATIC_FRICTION_ZERO, STATIC_FRICTION_MAX

Independent transfer checks

T01 - A spacecraft coasts left at constant velocity in deep space with approximately zero net external force. What happens to its velocity? Explain in one sentence.

Expected reasoning: It continues left at constant velocity; zero net force means zero acceleration.

T02 - A 5.0 kg cart feels 18 N right and 8 N left. Find acceleration.

Expected reasoning: (18-8)/5 = 2.0 m/s^2 right.

T03 - Two skaters push each other. One is twice as massive as the other. Compare forces and accelerations during the push.

Expected reasoning: Interaction forces are equal in magnitude and opposite in direction; the lighter skater has twice the acceleration magnitude if their masses differ by a factor of two.

Tutor-side reporting boundaries

- Report exact evidence: selected answer, quoted group turn, and concept linked to it.
- Use "possible misconception" or "needs confirmation" for one observation.
- Do not automatically assign final grades or rank students.
- Allow the professor to correct, delete or reject an agent-generated interpretation.
- Do not include the instructor key in student-accessible retrieval or prompts.

Textbook alignment

Aligned with OpenStax University Physics, Volume 1, Chapter 5. Original questions, diagrams and explanation text. Not official university-course material. The source URLs are included in the course manifest.